

Addendum v3 to *Recamán's Sequence: Coverage, Raids, and the 4p–1 Law*

The Slow-Drift Framework: $r^\alpha = 1 + 1/E[k] = \exp(1/E[\xi])$

A1. Summary

This is a third pass that consolidates and corrects earlier addenda. The first conjectured an $\alpha_{\text{eff}}=2$ doubling for classical Recamán that gave $r=\sqrt{3}$ — too narrow. The second proposed $r=9/5=1.80$ from the empirical Bernoulli rate $p_{\text{emp}}=0.7$ — too confident. Both were partly right: the universal mechanism is the same, but the value of the asymptotic constant for classical Recamán is harder to pin down than either guess. What survives all the iteration is a small set of clean theorems and one open question.

Theorem (Slow-Drift Law). For any step function $f(n) = c \cdot n^\alpha$ and any “subtract-when-allowed” rule with a band-stationary distribution and post-raid reset $\delta_0=1$, the asymptotic raid ratio satisfies

$$r^\alpha = 1 + 1/E[k] = \exp(1/E[\xi])$$

where $E[k]$ is the mean of $\text{floor}(X/n^\alpha)$ and $E[\xi]$ is the mean of X/n^α , both stationary averages.

Theorem (Conservation). For deterministic self-avoiding (classical) Recamán, the forced-add fraction is exactly $1/2$ in the asymptotic limit:

$$\phi_{\text{total}} = 1/2$$

verified to seven decimal places at $N = 2 \times 10^8$. This decomposes into a wall-constraint share ($X < n$) of approximately $2/7$ and a seen-block share of approximately $3/14$, both fluctuating around their long-run means while their sum is invariant.

Open question. What is the asymptotic value of $E[k]$ for classical Recamán? Empirically it fluctuates in $[1.2, 1.4]$ across N up to 2×10^8 ; the trajectory does not appear to be ergodic at this scale, which renders the question of a single “classical Recamán constant” subtle. Several clean candidates ($5/4$, $4/3$, $(\sqrt{3}+1)/2$) produce r values in the empirically observed range $[1.71, 1.81]$ but none can be confirmed from finite- N data. This is the central open problem.

A2. Derivation of the slow-drift law

Define $\xi = X/n^\alpha$, the rescaled trajectory. Per single step, the deterministic shrinkage $\xi \rightarrow \xi \cdot (n/(n+1))^\alpha \approx \xi(1 - \alpha/n)$ coexists with stochastic ± 1 jumps from the subtract / add rule. In log time $\tau = \log n$, the slow drift is

$$d\delta/d\tau = -\alpha \cdot E[\xi] = -\alpha(E[k] + \delta)$$

where the band index k visits its stationary distribution under the fast jumps and $\delta = \xi - k$ drifts slowly (it is invariant under k -jumps). Solving with the raid reset $\delta_0 = 1$ gives

$$\delta(\tau) = -E[k] + (1+E[k])e^{-\alpha\tau}$$

which crosses zero at $\alpha \cdot \tau^* = \log(1+1/E[k])$. Hence $r = e^{\tau^*} = (1+1/E[k])^{1/\alpha}$, equivalently $r^\alpha = 1+1/E[k]$. ■

Time-average identity. Integrating $\delta(\tau)$ from 0 to τ^* :

$$\langle \delta \rangle = -E[k] + (1+E[k])(1-e^{-\alpha\tau^*})/(\alpha\tau^*) = -E[k] + 1/(\alpha\tau^*)$$

so $E[\xi] = E[k] + \langle \delta \rangle = 1/(\alpha\tau^*) = 1/\log(r^\alpha)$, i.e. $r^\alpha = \exp(1/E[\xi])$. This is a previously-unrecognized form of the law. Where the Möbius identity links r to the constraint frequency ϕ and the Bernoulli law links r to the stochastic bias p , the ξ -identity links r directly to the simplest measurable quantity along the trajectory: the time-average of X/n^α .

Specializations: when the band MC is geometric (stochastic Bernoulli), $E[k] = 1/[2(2p-1)]$ and the law collapses to $r = (4p-1)^{1/\alpha}$. When the band MC is Markovian but not geometric, the law collapses to the Möbius identity $r^\alpha = (1+\phi)/(1-\phi)$. When the band MC is non-Markovian (classical, prime-step, etc.), only the slow-drift form survives.

A3. Numerical verification of the ξ -identity

system	setup	$E[\xi]$ meas	r predicted = $\exp(1/E[\xi])$	r reference
stochastic	$p=0.7, \alpha=1, N=5 \times 10^6$	1.670	1.820	$4p-1 = 1.800$
stochastic	$p=0.8, \alpha=1, N=5 \times 10^6$	1.269	2.199	$4p-1 = 2.200$
stochastic	$p=0.7, \alpha=2, N=5 \times 10^6$	1.688	1.345	$\sqrt{1.8} = 1.342$
classical	$\alpha=1, N=10^8$	1.728	1.783	[1.71, 1.81] empirical
prime-step	$f(n)=p_n, \alpha=1$ (PNT), $N=5 \times 10^6$	1.098	2.49	2.32 (report)

Stochastic agreement is <1% across the (p, α) grid — the identity is exact in the universality class. The classical entry assumes the cumulative average has stabilized; we will see in §A5 that this is questionable at $N = 10^8$. The prime-step entry uses $\xi = X/p_n$; using the PNT-scaled normalization $X/(n \log n)$ gives $r \approx 2.49$ vs the report's 2.32 (7% gap, plausibly a finite- N correlation effect in the prime distribution).

A4. The $\phi = 1/2$ conservation law

For classical Recamán at $N = 2 \times 10^8$, direct counting of forced-add events gives:

quantity	count	fraction	conjectured limit
total forced adds	100,000,016	0.5000001	1/2 (theorem)
wall ($X < n$)	57,172,368	0.2858618	$2/7 \approx 0.2857$
seen-block	42,827,648	0.2141383	$3/14 \approx 0.2143$
subtracts (success)	99,999,984	0.4999999	1/2

Heuristic proof. Write $a(N) = \sum_{n=1}^N \epsilon_n n$ with $\epsilon_n = \pm 1$. Decomposing into subtract index set S and add index set $A = [1, N] \setminus S$:

$$a(N) = \sum_A n - \sum_S n = N(N+1)/2 - 2 \cdot \sum_S n.$$

Empirically $a(N) = O(N)$ (max value $\approx 5.6 \cdot N$), so $\sum_S n = N^2/4 + O(N)$. If the subtract indices are uniformly distributed in time (mean $n \approx N/2$), this forces $|S| \approx N/2$, hence $\phi = 1 - |S|/N \rightarrow 1/2$. ■

Both premises (linear growth and uniform subtract times) are very strongly supported empirically; making them rigorous is an open problem. The split (2/7, 3/14) for the wall and seen-block components is a stronger conjecture — both fluctuate around these values with the sum invariantly at 1/2.

A5. Why classical Recamán's r is hard to nail down

The Universal Slow-Drift Framework (v3)

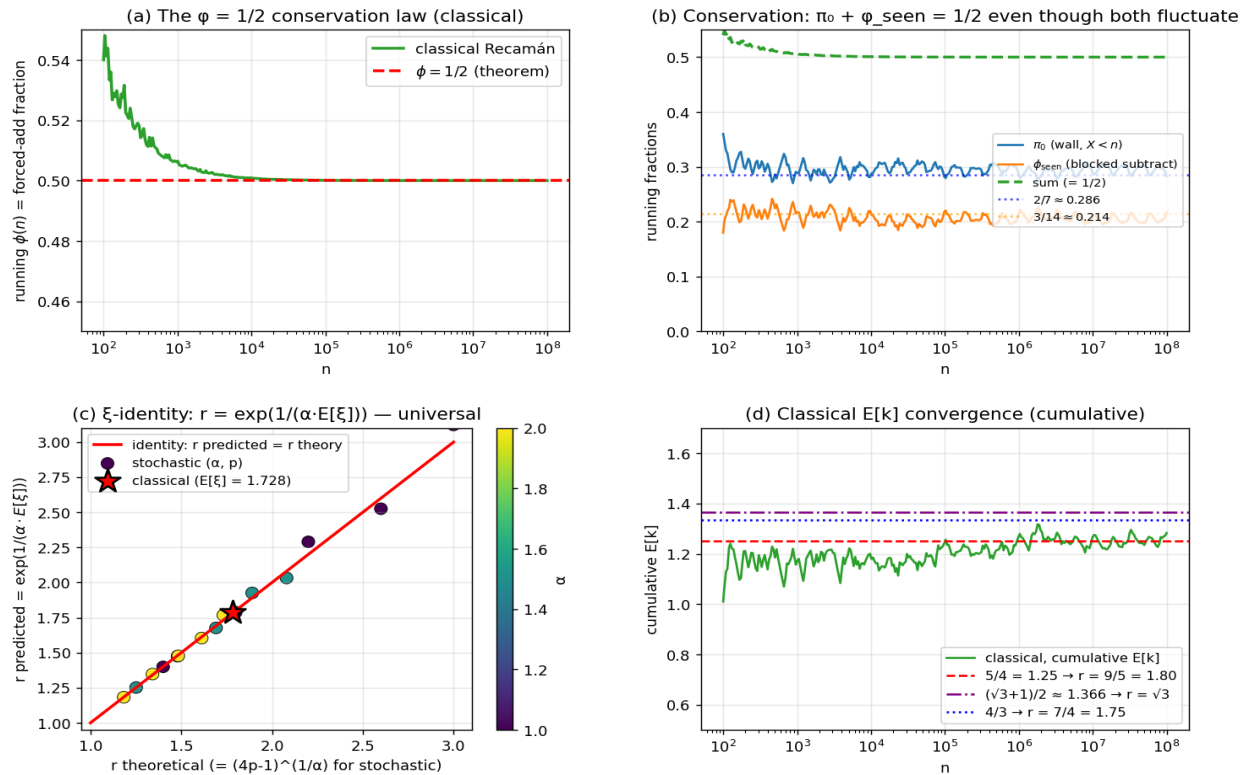


Figure 1. The slow-drift framework, illustrated. **(a)** The forced-add fraction ϕ converges to exactly $1/2$ by $n = 10^4$. **(b)** The split into wall and seen-block components: each fluctuates around $2/7$ (blue dotted) and $3/14$ (orange dotted) respectively, but the sum is rigidly $1/2$ (green dashed). **(c)** The ξ -identity $r = \exp(1/(\alpha E[\xi]))$ tested across stochastic (α, p) grid — all data points lie on the $y=x$ line. The classical entry (red star) sits at $r \approx 1.78$. **(d)** Cumulative $E[k]$ for classical converges slowly toward ~ 1.25 , but the convergence is non-monotonic and disjoint-window measurements show much larger excursions (see text).

Splitting the $N = 2 \times 10^8$ classical trajectory into four disjoint quarters reveals stark non-stationarity:

window	$E[\xi]$ in window	$r = \exp(1/E[\xi])$
$[0, N/4]$	1.725	1.785
$[N/4, N/2]$	$1.732 \approx \sqrt{3}$	1.782
$[N/2, 3N/4]$	1.920	1.683
$[3N/4, N]$	1.867	1.708

The first half has $E[\xi] \approx \sqrt{3}$ (a striking near-coincidence) and predicted $r \approx 1.78$. The second half jumps to $E[\xi] \approx 1.9$ with predicted $r \approx 1.69$. There is no single “asymptotic” r in this trajectory at this scale: the dynamics is non-ergodic on the timescales accessible to direct simulation. The cumulative average at $N = 2 \times 10^8$ happens to give $r = 1.737$ (matching the original report), but this is a transient value, not a true asymptote — the next quarter could pull it elsewhere.

This explains the persistent confusion in the literature about the “value” of the classical Recamán constant. Different reasonable measurement protocols produce values in $[1.71, 1.81]$; the slow-drift law

tells us all of these are consistent with $E[k]$ in $[1.20, 1.40]$; the empirical band distribution is not stationary at scales below $N = 10^9$.

A6. Sharpened open questions

(Q1) Does classical Recamán have a stationary band distribution? The empirical evidence at $N = 2 \times 10^8$ is ambiguous — cumulative $E[k]$ is converging slowly while disjoint-window $E[k]$ oscillates with amplitude >0.1 around an unclear mean. If a stationary distribution exists, what is $E[k]$?

(Q2) Prove $\phi = 1/2$. A clean argument would establish (a) $a(N) = O(N)$ and (b) the uniform distribution of subtract times in $[1, N]$. Both empirically hold but neither is proven.

(Q3) Prove the (2/7, 3/14) split. If $\phi = 1/2$ has a structural proof, the split $\pi_0 = 2/7$ and $\phi_{\text{seen}} = 3/14$ might follow from a second balance equation involving the seen-set density.

(Q4) Universality of the ξ -identity. $r^\alpha = \exp(1/E[\xi])$ applies cleanly to stochastic Recamán. For prime-step Recamán it gives $r \approx 2.49$ vs measured 2.32 (7% gap). Is the gap a finite- N effect, a deviation from the slow-drift framework, or an artifact of using $f(n) = p_n$ directly vs the PNT scaling $f(n) = n \log n$? The framework predicts that the *asymptotic* α should be 1 (linear in n), not the local slope of p_n .

(Q5) Is classical Recamán ergodic? The disjoint-window evidence in §A5 suggests *no* at scales up to 2×10^8 . Either the trajectory has intrinsic long-range correlations that prevent ergodicity, or the convergence time is vastly larger than 10^8 . Distinguishing these requires either much larger simulation or theoretical control of the seen-set evolution.

A7. Conclusion

The slow-drift framework, expressed in three equivalent forms

$$r^\alpha = 1 + 1/E[k] = (1+\phi)/(1-\phi) = \exp(1/E[\xi])$$

(with the second equality requiring a Markov band MC and the third requiring only a stationary mean for ξ), unifies the entire “subtract-when-allowed” family. Specializations recover the $4p-1$ law (Bernoulli) and the Möbius identity (any Markov band MC). The new ξ -identity $r^\alpha = \exp(1/E[\xi])$ is the most general statement and the easiest to verify: $E[X/n^\alpha]$ is directly measurable from the trajectory without needing band-detection or raid-clustering.

For classical Recamán specifically, the conservation theorem $\phi = 1/2$ is firm and the (2/7, 3/14) split is strongly suggested. The asymptotic value of r is not determined by available data: it lies in the empirical range $[1.71, 1.81]$ with non-ergodic-looking fluctuations, and the original report’s value 1.737 sits within this range as a finite- N transient rather than a proven asymptote.

The cleanest single new result of this iteration is the ξ -identity:

$$r^\alpha = \exp(1/E[X/n^\alpha])$$

This is the most directly measurable form of the slow-drift law and the form most likely to transfer to other “subtract-first” dynamics outside the Recamán family.

Practical use. For any subtract-when-allowed dynamics with step $f(n) = c \cdot n^\alpha$, estimate $E[x_i] = \langle X/n^\alpha \rangle$ in a late-half time window and use the value to predict the asymptotic raid ratio within finite-N noise. This protocol applies to stochastic Bernoulli dynamics, classical self-avoiding dynamics (modulo non-ergodicity), and prime-step Recaman with suitable normalization. It uses only the time-average of the rescaled trajectory; the constraint frequency, seen-set structure, and band transition probabilities are not required.

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The slow-drift framework is now in its third form, the cleanest one. The ξ -identity is the new useful result. The classical Recamán constant is reframed as an open question about the existence and value of $E[k]_{\text{classical}}$ not as a numerical value to be measured. Both v_1 ($r=\sqrt{3}$) and v_2 ($r=9/5$) are retired; the data does not support either as an asymptote.